

Gahyun Yoo

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Education

MS	University of Southern California , Computer Science	Aug 2026 – July 2028
BS	Seoul National University , Linguistics & Computer Science	Mar 2020 – Aug 2025

Experience

DeepAuto , Agentic AI Engineer (Freelance)	Apr 2026 – June 2026 3 months
- Major conglomerate EPC automation project - Proposed & developed data lakehouse system for the EPC domain comprised of a tri-store knowledge base - Developed lakehouse-connected agents that each boosted recall for instrumentation drawing retrieval by 10%pp - Containerization for efficient and reliable forward deployment - Improved object detection by referencing project-specific legend sheets	
SKI-ML Lab (PI: Jay-Yoon Lee) , Seoul National University , Research Assistant	Sept 2025 – Mar 2026 7 months
[Research Project] Reward Model Finetuning for Diffusion Model Alignment Using Tree Structured Trajectories [Government Project] Development of a Korean-English Translation Model for Safety Management of Pharmaceutical Products (Ministry of Food and Drug Safety): As project lead, developed a machine translation pipeline supporting an ensemble of inference strategies using in-context examples and dictionary entries, and instruction tuning & preference optimization training, outperforming several SOTA commercial models [TA] Special Topic: Multimodal & Diffusion Models [Coursework] Special Topic: Generative Modeling with Diffusion and Flow-Based Models	
SKI-ML Lab (PI: Jay-Yoon Lee) , Seoul National University , Research Intern	Mar 2023 – Aug 2025 2 years 6 months
[Research Project] Application of energy based training with quality estimators as score models for low resource machine translation, resulted in a publication at COLING 2025 [Research Project] Advised machine translation project, resulting in publication at ACL 2025	
NCSOFT NLP Center , Research Intern	July 2022 – Aug 2022 2 months
Developed a decision tree classification algorithm for distinguishing literal and non-literal translation, using linguistic features, alignment score from text embedding similarity, and multilingual knowledge graphs (>90% accuracy on open source English-Chinese dataset)	

Publications

Improving NMT Models by Retrofitting Quality Estimators into Trainable Energy Loss Gahyun Yoo, Jay-Yoon Lee (International Conference on Computational Linguistics)	Jan 2025
BridG MT: Enhancing LLMs' Machine Translation Capabilities with Sentence Bridging and Gradual MT Seung-Woo Choi, Gahyun Yoo, Jay-Yoon Lee (Findings of the Association for Computational Linguistics)	July 2025

Generalizing Numerical Reasoning in Table Data through Operation Sketches and Self-Supervised Learning

May 2026

Hanjun Cho, *Gahyun Yoo*, Hanseong Kim, Jay-Yoon Lee

(Transactions of the Association for Computational Linguistics)

Awards

EFL 8th Best Speaker, SNUDA X KNC 2021

Korea Intervarsity Debate
Association
Jan 2021

Bronze Medal, Korean Mathematical Olympiad

Korean Mathematical Soci-
ety
Jan 2014

Presentations

- Invited Seminar @ Kakao Mobility (Feb 2025)

Extracurricular Activities

AttentionX: Member of Artificial Intelligence Research Group (2023)

Waffle Studio: Member of Seoul National University Web Development Club (2021-2023), developed the backend server of an interactive web editor application

SNUDA: Finance Minister and Member of Seoul National University Debate Club (2020-2021)

Skills

Languages: Python, C++, Java

ML Frameworks: PyTorch, SGLang, VLLM, Tensorflow, Scikit-Learn

Infrastructure: Docker

Coursework

Math: Linear Algebra, Calculus 1&2, Statistical Computing and Lab, Mathematical Analysis, Discrete Mathematics

Computer Science: Data Structures, Algorithms, Automata Theory, Logic Design, Computer Architecture, System Programming

Artificial Intelligence: Basics of Deep Learning, Machine Learning and Deep Learning for Data Science 1&2, 'Special Topic - Generative Modeling with Diffusion and Flow-Based Models'